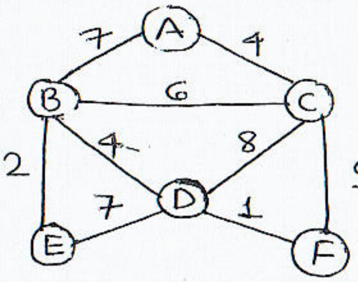
12.12.2019 (E)

K. J. Somaiya College of Engineering, Mumbai-77 **(Autonomous College Affiliated to University of Mumbai)**
End Semester Examinations Nov - Dec 2019
Max. Marks: 100 Class: SY BTech Name of the Course: Analysis of Algorithms Course Code :UITC403
Duration: 3 hrs Semester: IV Branch: IT
Instructions: **(1) Question 1 Compulsory** **(2) Out of remaining solve any 4.** **(3) Draw neat diagrams** **(4) Assume suitable data if necessary**

Question No.		Marks
Q 1 (a)	How would you calculate time complexity of the following recurrence relationship using recurrence tree? $T(N)=T(N/3)+T(2N/3)+O(N)$	10
Q 1 (b)	Write algorithm for binary search. Derive its complexity.	10
Q2 (a)	Explain N-queens problem with example.	10
Q2 (b)	Explain time complexity. Compare time complexities of bubble, heap, radix, shell sort.	10
Q3 (a)	Sort the following numbers using quick sort. Also derive the time complexity of Quick sort. 17, 2, 15, 18, 20, 1, 9, 8 11, 4	10
Q3 (b)	Explain Strassen's matrix multiplication.	10
Q4 (a)	Find minimum spanning tree of the following graph using Kruskal's algorithm.	10



Q4 (b)	Find solution of given knapsack problem using greedy method. N=4, M=5 W1,W2,W3,W4= 1,3,2,5 P1,P2,P3,P4 = 200,240,140,150 Where w is weight and p is profit	10

Q5 (a)	Write and Explain sum of subset algorithm for n=5, W={2,7,8,9,15} m=17.	10
Q5 (b)	Write short note on:(solve any two) a)Job sequencing algorithm b) Branch & Bound method c) Randomized algorithm.	10
Q6 (a)	Write short note on Optimal Binary Search Tree using dynamic programming	10
Q6 (b)	Explain Travelling Salesman Problem. With a suitable example, explain how TSP can be solved with Greedy approach	10
Q7 (a)	How would you analyze time complexity of multiplying long integer?(mention algorithm)	10
Q7 (b)	Explain KMP algorithm with example.	10